4 · *From Niche to Distribution: Basic Modeling Principles and Applications*

In this chapter, we describe and illustrate the principles for modeling the distribution of suitable habitats for a given species. However, as discussed in Chapter 1, habitats for other biological entities can also be modeled using the same approach, these may include intraspecific levels (e.g. subspecies, haplotypes), supra-specific levels (e.g. functional groups, communities, ecosystem types), or features that are transversal across species (e.g. species traits, genes or alleles, etc.).

Here, we address how to fit the niche of the modeled entity from field observations (4.1), how the fitted species' niche can then be projected into geographical space to predict species distributions (4.2), how individual single species predictions can be assembled into community-level predictions (e.g. species richness; 4.3), and finally the possible applications of these models and their predictions in ecology, biogeography and evolution (4.4).

4.1 From Geographical Distribution to Niche Quantification

We saw in Chapter 3 that each site on Earth is characterized by a set of environmental conditions, which define a specific habitat inhabited, or not, by a community of species (Kearney, 2006). Habitats are sometime characterized *per se*, as distinguishable units (e.g. discrete vegetation units), but most often they are simply described based on a set of variables that express various characteristics of the environment, such as climate, land use, and soil for plants, or climate, food resources, and habitat structure for animals, at one or a series of locations. Quantifying a species' realized environmental niche (see 3.3 and 3.4), makes it possible to assess the habitat suitability of any geographic unit for which those environmental niche characteristics are known, i.e.

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an environmental variable with a spatially explicit coverage, as obtained through and managed in geographic information systems (GIS) and remote-sensing (RS) technologies (see Part II). Once spatially explicit layers (GIS maps) are available for relevant environmental variables defining a species' niche, it is possible to predict the potential distribution of the species' suitable habitats within the study area. This is the basic principle of habitat suitability modeling (or niche-based species distribution modeling if one can expect the whole niche envelope to be captured within the study area). HSMs thus predict a snapshot of the realized environmental niche for a species in geographic space for a given period of time (i.e. defined by the sampling dates and the environmental variables used).

How is the species' realized niche quantified? One simple approach is to take all known localities of the species and for each of them extract the attributes of the environmental variables that define the species' niche (Figure 4.1a). The next step is to relate the species observations to the environmental data in order to obtain a model of the species–environment relationship. When variation in abundances and/or known absences of the species at localities are also available, they can greatly improve the discrimination between suitable and unsuitable habitats, and therefore improve the quantification of the niche (Brotons et al., 2004; Howard et al., 2014), but such model improvement may not always be observed (Pearce and Ferrier, 2001). Such species' occurrence, presence–absence or abundance data are usually related to the corresponding environmental data using simple rules (e.g. expert or logical) or more advanced statistical techniques (Figure 4.1b; see Section 4.4 and Part III). However, many observations of species distributions used to calibrate HSMs and predict species distributions are simple occurrences, i.e. observations of species presence with no information of absence, often originating from heterogeneous databases of natural history collections (Graham et al., 2004a; see Chapter 7). In the best case scenario, these might be presence–absence data from an atlas or designed scientific surveys, or even abundance data or population fitness measurements, but these scenarios are surprisingly infrequent in the HSM literature. Observations are thus most commonly based on simple observations, and are rarely associated with field measurements of abundance or population fitness, and thus do not provide any measure of these. Without population fitness measurements, such data can potentially reflect some of the suboptimal sink situations described in Chapter 3, which might in turn affect the habitat suitability and niche estimations (Pulliam, 2000).

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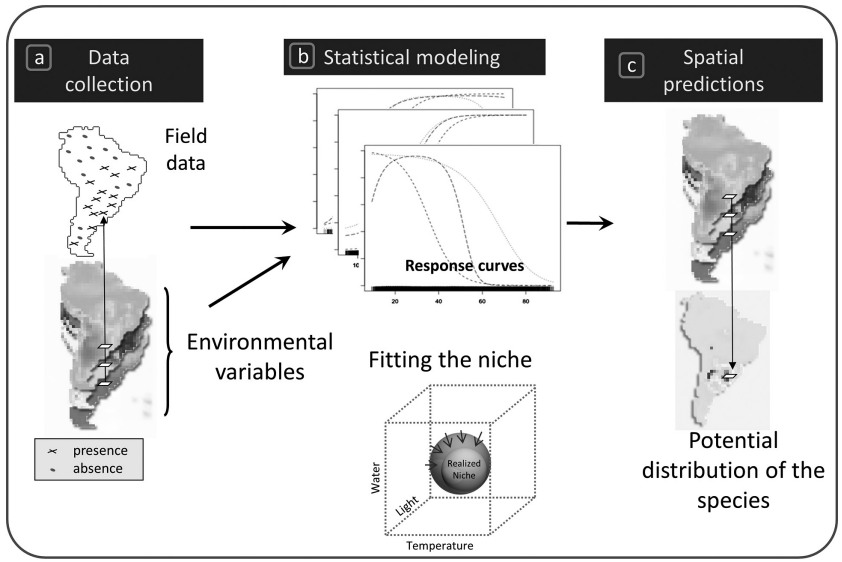


Figure 4.1 Principle of habitat suitability modeling, illustrated on a study area representing South America. (a) Field observations are collected and geo-referenced and the attributes of a set of environmental maps covering the area are extracted for each of them (see Part II). (b) Species observations are related to the environmental values using statistical approaches (Part III) to fit species response curves to each environmental predictor, in order to quantify the envelope of suitable habitat conditions, which represents the realized environmental niche of the species or of part of it if the study area does not encompass all the environmental conditions suitable for the species (see Chapter 3; the arrows suggest competitive exclusion of part of the fundamental niche from other species). (c) The fitted habitat suitability model is then used to combine the initial environmental maps and come up with a spatial projection of the model in geographic space (Part V), corresponding to the species' potential distribution. (*A black and white version of this figure will appear in some formats. For the color version, please refer to the plate section.*)

Whether absences are used or not is an important question because, as we will see later (Part III), it can condition which modeling technique is used to fit and predict the niche. However, any useful information is important when modeling species distribution. So, if absence data, for example, are available and reliable then they should be used. The modeling techniques used to fit the niche may differ (Part III). The niche may be simply captured using a series of Boolean rules (e.g. min and max along gradients, see Figure 4.2a), but in most cases it is based on statistical functions (Figure 4.2b; see also Part III). In general, the functions describing the species distribution along each environmental variable

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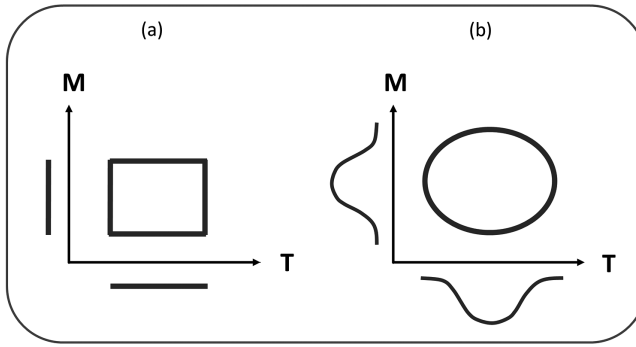


Figure 4.2 Illustration of how modeled species' responses to environmental predictors (e.g. temperature T and moisture M) in the form of simple ranges of tolerance lead to "box-like" (e.g. BIOCLIM; see Part III) representations of the niche (a), whereas smoothed unimodal response curves lead to more realistic ellipse-like niche representations (b; here using a minimum value of the combined responses to define the envelope, e.g. at predicted probability = 0.1; see Part III).

in these models are called "species response curves" (Austin et al., 1994). These response curves therefore either represent simple rectilinear "box-like" envelopes resulting in simple binary (inside/outside the niche) predictions (Figure 4.2a), or more realistic representations of the niche (e.g. unimodal) based on more gradual (and often more complex) responses in environmental space (Figure 4.2b), resulting in continuous index of habitat suitability or probability of species occurrence (usually from low to high, e.g. [0–1] or [0–100]; see Part III and Merow et al. (2014) for discussion of simple versus complex response curves and associated models).

A crucial step in habitat suitability modeling is the acquisition of spatially explicit environmental variables (i.e. maps) at the right resolution, which are sufficiently accurate to be used to determine a species niche as close to its ecophysiological needs as possible. As previously seen (3.3), environmental variables (or predictors; see Part II) can exert direct or indirect effects on species, which can be expressed as a gradient ranging from proximal to distal predictors (Austin, 2002; Huston, 2002; Austin, 2007; Mod et al., 2016). These are ideally chosen to reflect the three main types of influences on the species (Guisan and Zimmermann, 2000; Guisan and Thuiller, 2005): (i) regulators (or limiting factors), defined as factors controlling a species' metabolism (e.g. low temperatures); (ii) disturbances, defined as all types of perturbations

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affecting environmental systems (natural or human-induced); and (iii) resources, defined as all compounds that can be consumed by organisms (e.g. radiations for plants, water, prey for animals, nutrients for plants). All these variables are considered to be proximal predictors and are the preferential option when building HSMs. However, there are many other variables available that exert an indirect, rather than a direct, effect on species distribution. One extensively discussed example of an indirect (or distal) predictor is elevation (see Körner, 2007), which indirectly affects species distributions through its impact on numerous co-varying variables, such as solar radiation, mean annual temperature, precipitation, or topography (e.g. frequency of steep slopes). Unfortunately, the correlation with these more proximal variables is not constant across the globe, and therefore elevation (or other indirect variables) can only produce meaningful predictions when applied to relatively small regions where the conditions are unchanging (no climate or land-use change, for example).

In practice, the environmental variables available as spatial layers or maps usually vary in their degree of proximality (Austin, 2002, 2007; Mod et al., 2016). Furthermore, it can be difficult to evaluate how proximal a variable really is, because its effects are species specific (e.g. not all species are limited by low temperatures) and physiological causality can only be assessed experimentally (available only for a limited number of species). Therefore, habitat suitability modeling largely depends on prior ecological and ecophysiological knowledge, be it experimental or theoretical, about factors that physiologically and ecologically determine species distributions (e.g. Woodward, 1987; Jones, 1992 for plants). The more detailed our knowledge of the physiological drivers of species distributions, the more precisely we can identify what types of environmental maps are necessary, which will ultimately make our predictive ability more accurate.

As the most proximal variables (e.g. soil physical and chemical properties for plants) are often not available, at least in a spatially explicit way, the most common solution is to use topographic descriptors (least proximal) or environmental variables that represent averages or sums of daily or monthly values (e.g. annual mean air temperature, growing degree days; see Chapter 5) that are biologically meaningful and thus expected to provide a sufficiently good approximation of the true proximal variables (Mod et al., 2016). The degree of proximality of the predictor variables used to quantify the niche thus also conditions the degree of

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accuracy that can be expected when using the resulting model to make predictions (Chapter 5). However, this does also depend on the scale of the study (Austin and Van Niel, 2011).

Therefore, because of its correlative nature and due to the lack of proximality of some predictors, a statistical approach to niche quantification and habitat suitability modeling can only support previously evidenced biogeographical theories and hypotheses, or suggest new ones. It does not constitute definitive proof of a theory in its own right. However, it does provide a simple and efficient way of making biogeographic predictions (see below).

4.2 From the Quantified Niche to Spatial Predictions

After fitting the realized environmental niche (Figure 4.1b), or part of it (i.e. any envelope of suitable habitats) in a model, the next question we need to ask is: how can the quantified niche be projected into geographic space to obtain the potential distribution of a species (Figure 4.1c)? In order to answer this question, we need to transfer the statistical model, which contains a quantification of a species' response in environmental space, back to the geographic space in which the species was originally observed. To do this, one can apply the statistical functions, fitted in environmental space, to any position in geographic space using the environmental conditions at these locations from the corresponding maps. This makes it possible to express the suitability of the habitat for the species modeled in a spatially explicit way across the whole study area (Figure 4.1c). For spatial predictions, we therefore need to assign a habitat suitability value to every geographic cell in the area by applying the models to the spatial layers representing the niche variables used in the model. Since all the approaches to quantifying a species' niche result in these types of envelopes or more refined niche quantifications in environmental space, they all can yield spatial predictions. However, the exact way of doing it can differ enormously and depends on the modeling technique used (Parts III and V).

We will see later, in Part V, that many statistical models include a function for predicting a suitability value at any new location using the environmental variables available for this location. The most straightforward way of predicting the species' habitat suitability is to use this function to make predictions for all cells of the study area. To do this the function needs to be able to use a pile (or stack) of environmental maps

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corresponding to the variables used as predictors in the HSM (Figure 4.1c). This will yield a final habitat suitability value for each cell. If the environmental values for a set of species presence–absence locations are first entered into a statistical model as the input and then directly predicted back to a stack of environmental layers, without visualization in environmental space, then we can consider this to be a blind approach. A better way of understanding how a species' niche response has been fitted in a model is thus to first check the prediction process in environmental as opposed to geographic space alone. This can be done by projecting the habitat suitability into the environmental space in which the niche was originally fitted.

So far, we have discussed cases where models have been used to make predictions in the same area and under the same environmental conditions as those used to fit the models. However, there are many instances where predictions need to be made for separate areas or for different time periods, potentially with different environmental conditions, thus "projecting" the model and niche in space and time (Pearson and Dawson, 2003; Heikkinen et al., 2006; Pearman et al., 2008a; Thuiller et al., 2008; Elith et al., 2010; Bellard et al., 2012). Projecting a model to a different area can, for instance, be useful when assessing the invasive potential of an exotic species on a new continent, with the model being fitted in the native range of species and then projected worldwide or to specific areas (Peterson, 2003; Thuiller et al., 2005b; Broennimann et al., 2007; Fitzpatrick et al., 2007; Petitpierre et al., 2012; Guisan et al., 2014). Being able to project a model to another time period can, for instance, be useful when assessing the potential impact of future climate and land-use changes on species (Iverson and Prasad, 1998; Peterson et al., 2002a; Meyneeke, 2004; Simmons et al., 2004; Thomas et al., 2004; Thuiller et al., 2005a; Araújo et al., 2006; Engler et al., 2011b) and on biodiversity (Buisson et al., 2008; Thuiller et al., 2011; Pio et al., 2014; Thuiller et al., 2014a; Thuiller et al., 2014c). We can also assess our capacity to predict the future by testing how successfully we can project models back in time or from the past to the present (Araújo et al., 2005a; Pearman et al., 2008b; Maiorano et al., 2013), or couple HSM projections to the past using phylogeographic or phylogenetic studies (e.g. Hugall et al., 2002; Carnaval et al., 2009; Schorr et al., 2013).

How these models can be used to derive predictions under scenarios of climate and/or land-use change will be explained and treated in

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greater depth in Part V. As we will see, it is actually done in the same way as for predicting a model under current environmental conditions in the same area, but the model is applied to a distinct set of input environmental maps (see Part V), either for a distinct study area (for instance, the invaded range for an exotic species) or for a distinct time period according to certain environmental change scenarios (e.g. in land use or climate).

4.3 From Individual Species Predictions to Communities

We have introduced how spatial predictions can be made for individual species, under current or future conditions, or in one or several study areas. However, in many instances, several species are modeled simultaneously and the next step is to infer community or diversity patterns from the individual species models. Here, we discuss how this can be done.

Numerous studies to date have looked at changes in species composition (i.e. species turnover) in space or time. For example, measurements of species turnover are often generated in studies assessing future threats from climate change (e.g. Thuiller et al., 2005a; Engler et al., 2011b). These metrics are comparable to those used to measure changes in community composition (i.e. similarity coefficients; e.g. Soerenson) or richness (beta-diversity) between survey units (see Legendre et al., 2005). Such metrics usually compare the number and identity of shared species to the total pool of species between the two communities or regions being contrasted. It is thus a measure of species difference. Caution should therefore be taken and explanations given when applying such measurements to surveys with partial taxonomic coverage, as is the case when using incomplete atlas data (e.g. *Atlas Florae Europaeae*, Jalas and Suominen, 1972–1996; Thuiller et al., 2005a).

In cases where species are exhaustively surveyed for one taxonomic group within spatial units small enough to still correspond to a meaningful community (e.g. small pixels, field quadrats; their upper meaningful size will depend on the group studied), modeling a large set of species opens up perspective for community analyses based on assembling single species predictions (Ferrier and Guisan, 2006). By cumulating single species predictions (i.e. stacked species distribution models, S-SDMs;

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Dubuis et al., 2011), one can attempt to reconstruct patterns of species richness and community properties. This is an emerging field (Ferrier and Guisan, 2006; Guisan and Rahbek, 2011; Kissling et al., 2012; Wisz et al., 2013; D'Amen et al., 2015b) and there are therefore still relatively few examples available (e.g. Guisan and Theurillat, 2000; Iverson and Prasad, 2001; Peppler-Lisbach and Schroder, 2004; Parviainen et al., 2009; Thuiller et al., 2011; Civantos et al., 2012; Mateo et al., 2012; Buisson et al., 2013; Pottier et al., 2013; Thuiller et al., 2014a; Thuiller et al., 2014c; D'Amen et al., 2015a).

The important question regarding community assembly is: to what extent can single species models integrate biotic interactions, site history and human disturbances that condition species assemblages, or how many additional constraints need to be included in a formal assembly process, for instance to limit the number of species possibly co-occurring in a given modeled unit from the potential pool of species predicted to occur there by the HSMs (Guisan and Rahbek, 2011; D'Amen et al., 2015a). This "spatially explicit species assemblage modeling" (SESAM; Guisan and Rahbek, 2011; Pottier et al., 2013; D'Amen et al., 2015a) approach provides a way of testing theories and hypotheses about how communities assemble. It opens up additional perspectives for HSM applications to address basic and applied questions in community and ecosystems ecology.

4.4 Main Fields of Application

In recent years, predictive modeling of species distribution based on estimating habitat suitability (HSMs) has become an increasingly important tool for addressing a range of issues in ecology, biogeography, evolutionary biology, conservation biology, and climate change impact research. HSMs can be used to answer both fundamental and applied questions, such as identifying the drivers of species distributions, testing biogeographic hypotheses, assessing niche conservatism, anticipating biological invasions, or assessing global change impacts on species distribution and diversity. They are also increasingly applied to entities transversal to species, such as functional traits, infraspecific taxa, or genes, and they are now coupled with other tools and data from other disciplines, in order to answer questions in phylogeography, phylogenetics, population genetics, population dynamics, and other fields.

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Table 4.1 *Some possible uses and applications of HSMs.*

Type of use	Examples of studies
1. Quantifying the environmental niche of species and its changes in time and space	Austin et al. (1990); Westman (1991); Vetaas (2002); Wharton and Kriticos (2004); Luoto et al, (2006); Broennimann et al. (2012); Guisan et al. (2014)
2. Disentangling the environmental drivers determining species ranges	Duckworth et al. (2000); Anderson et al. (2002); Leathwick (2002); Normand et al. (2009); Boulangeat et al. (2012a); Gallien et al. (2015)
3. Relating HSMs to species characteristics or population demography	Huntley et al. (2004); Thuiller et al. (2010); Kharouba et al. (2013); Thuiller et al. (2014b)
4. Testing evolutionary hypotheses in biogeography	Leathwick (1998); Anderson et al. (2002); Wiens and Graham (2005); Moloney and Jeltsch (2008); Evans et al. (2009); Boucher et al. (2012); Broennimann et al. (2014c)
5. Assessing species invasion and proliferation	Beerling et al. (1995); Dirnbock et al. (2003); Peterson (2003); Thuiller et al. (2005b); Peterson et al. (2008b); DeVaney et al. (2009); Gallagher et al. (2010); Gallien et al. (2012); Petitpierre et al. (2012); Broennimann et al. (2014a)
6. Assessing the impact of climate, land use and other environmental changes on species distributions	Beaumont and Hughes (2002); Peterson et al. (2002a); Meyneeke (2004); Simmons et al. (2004); Thomas et al. (2004); Thuiller et al. (2005a); Araújo et al. (2006); Broennimann et al. (2006); Pio et al. (2014); Thuiller et al. (2014a); Thuiller et al. (2014c)
7. Suggesting unsurveyed sites with a high potential of occurrence for rare or new species	Raxworthy et al. (2003); Engler et al. (2004); Edwards et al. (2005); Guisan et al. (2007b)
8. Supporting appropriate management plans for species recovery and mapping suitable sites for species restoration	Pearce and Lindenmayer (1998); Côté and Reynolds (2002); Regan et al. (2008)
9. Supporting conservation planning and reserve selection	Ferrier et al. (2002); Araújo et al. (2004); Kremen et al. (2008); Alagador et al. (2011); Meller et al. (2014)

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Table 4.1 (*cont.*)

Type of use	Examples of studies
10. Modeling community and ecosystem properties from individual species predictions	Guisan and Theurillat (2000); Ferrier et al. (2002); Olden (2003); Peppler-Lisbach and Schroder (2004); Algar et al. (2009); Guisan and Rahbek (2011); Pottier et al. (2013)
11. Detecting and anticipating disease spread and outbreaks	Peterson et al. (2002b); Peterson (2003, 2006); Estrada-Peña and Venzal (2007); Meentemeyer et al. (2008); Williams et al. (2008); de Oliveira et al. (2013); Yañez-Arenas et al. (2014); Zhu and Peterson (2014)
12. Incorporating habitat suitability into landscape (meta-)population dynamic assessments	Ferrier et al. (2002); Larson et al. (2004); Binzenhofer et al. (2005); Anderson et al. (2009); Brook et al. (2009); Dullinger et al. (2012); Naujokaitis-Lewis et al. (2013); Boulangeat et al. (2014)

As the number of papers on, and topics addressed using, HSMs has increased exponentially in recent decades (Guisan et al., 2013), it would be impossible to list them all. Instead, we have provided some examples in Table 4.1 and refer readers to review papers such as (Guisan and Zimmermann, 2000; Guisan and Thuiller, 2005; Thuiller et al., 2008; Elith and Leathwick, 2009; Zimmermann et al., 2010; Guisan et al., 2013; Thuiller et al., 2013) or complementary books (Scott et al., 2002; Franklin, 2010a; Peterson et al., 2011).

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